

What to build: three features, six tables, and what to pull out of each paper

Every table we need, every field, what it means in plain words, and whether we already have the data or still have to get it.

All the tables

Six tables. Two hold the questions, three record what a student did, one is a running total we keep so we don't recalculate it constantly.

CONTENT		WHAT A STUDENT DID	
papers	305 rows	attempts	one per sitting
directions	~3,000 rows	attempt_answers	one per answer
questions	18,651 rows	user_topic_stats	a running total

papers One row per exam paper		305 rows
FIELD	IN PLAIN WORDS	STATUS
paper_id	Which document this is. Everything hangs off it.	HAVE
bank	SBI, IBPS	93%
role	PO, Clerk, RRB	97%
exam_type	Prelims or Mains	94%
year	2019, 2022...	96%
shift	Morning / afternoon sitting	14%
memory_based	Recalled by students, not official	HAVE
exam_key	Which <i>real exam</i> this recalls. Several papers can describe the same exam.	BUILD
is_canonical	Show this one; hide the other recalls of the same exam.	BUILD
duration_min	How long the exam is. Needed for the timer.	NEED
total_marks	Marks for the whole paper.	NEED
section_timing	Prelims time each section separately.	NEED
source_pdf	Which PDF it came from, for tracing mistakes.	HAVE
is_active	Hide a bad paper without deleting it.	BUILD

questions One row per question 18,651 rows		
FIELD	IN PLAIN WORDS	STATUS
q_id	This question's id.	HAVE
paper_id	Which paper it came from. Never drop this.	HAVE
q_num	Its number in that paper. Sets the order in a mock.	HAVE
stem	The question text.	HAVE
options	The choices a–e.	98%
answer	Which one is correct.	0%
explanation	Why that answer is right.	0%
section	Reasoning / Quant / English / GA	15%
topic	Finer than section — “Seating arrangement”.	0%
difficulty	1–5. Lets us balance a set.	NEED
direction_id	Which passage it belongs to, if any.	71%
marks	Marks for a correct answer. Usually 1.	NEED
negative_marks	Deducted for a wrong one. Usually 0.25.	NEED
content_hash	Fingerprint of the text, so the same question from two sources becomes one row.	BUILD
is_active	Retire a broken question without losing its history.	BUILD

directions Passages shared by several questions ~3,000 rows		
FIELD	IN PLAIN WORDS	STATUS
paper_id	Part of the key. Passage ids repeat across papers.	HAVE
direction_id	d001...d030, only unique inside one paper.	HAVE
body	The passage or instruction text.	HAVE

All 30 of our passage ids appear in many papers. Look one up by `direction_id` alone and the student reads a passage from a different exam. Always use `paper_id + direction_id` together.

attempts One row each time a student sits something		grows with use
FIELD	IN PLAIN WORDS	
attempt_id	This sitting.	
user_id	Who.	
mode	bank , mix or paper .	
paper_id	Set only for a full paper.	
started_at	When the timer began.	
submitted_at	Empty until they finish. Lets them resume.	
score	Final marks, after negatives.	

attempt_answers One row per question answered		the biggest table
FIELD	IN PLAIN WORDS	
attempt_id	Which sitting.	
q_id	Which question.	
chosen	What they picked, a–e. Empty if skipped.	
is_correct	Right or wrong.	
time_ms	How long they took. Shows where they lose time.	

This is the table that makes mastery, the progress counter and the grade rail possible. Without it we can show questions but can never say a student is improving.

user_topic_stats A running total per student per topic		derived
FIELD	IN PLAIN WORDS	
user_id · topic	Who, and which topic.	
attempted · correct	Counts, updated as they answer.	
accuracy	correct ÷ attempted. This is mastery.	
last_seen_at	Feeds spaced revision.	

Could be calculated from `attempt_answers` every time, but that gets slow. Keeping a running total means the dashboard is one quick read.

What to pull out of each paper

Run once per JSON file. Ten things.

FROM THE FILE – already there

- 1 paper_id use as-is
- 2 bank role e.g. SBI · P0
- 3 exam_type year Mains · 2019
- 4 shift usually missing, that's fine
- 5 source pdf path keep for tracing

WORK OUT OURSELVES – not in the file

- 6 exam_key bank + role + exam_type + year + shift
joined together. Papers sharing a key
describe the same real exam.
- 7 is_canonical of the papers sharing an exam_key, mark
the most complete one; hide the rest
- 8 duration_min from the exam pattern, not the file
(SBI P0 Mains = 180, Prelims = 60)
- 9 total_marks same – from the pattern
- 10 section_timing which sections, how long each

What to pull out of each question

Run once per question. Fifteen things, and four of them we don't have yet.

FROM THE FILE – already there

- 1 q_id
- 2 q_num its number in the paper
- 3 stem the question text
- 4 options a–e (98% of the time)
- 5 direction_id which passage, if any

WORK OUT OURSELVES – cheap, do it in the import script

- 6 content_hash hash of stem + options, to spot the same question arriving from two sources
- 7 marks 1, from the exam pattern
- 8 negative_marks 0.25, from the exam pattern
- 9 is_active false if options are missing or empty

STILL MISSING – real work, not a script

- 10 answer ← nothing works without this
- 11 explanation why the answer is right
- 12 section Reasoning / Quant / English / GA
- 13 topic "Seating arrangement", "Profit & loss"
- 14 difficulty 1–5

Items 10–14 are the whole gap. Everything else is a few lines of code in the import script. **Answer keys are the one that blocks two of the three features** — a timed test that cannot mark itself is not shippable.

The three features

Same questions in all three. What differs is how much time the student has.

01 Question Bank

2 minutes

Question Bank 12 / 240 done
SBI · P0 · Reasoning

Q. A, B, C, D and E sit in a row...
() 3rd right (•) 5th left
() 2nd left () 3rd left

[Check] Skip →

- One question at a time. Answer, told right or wrong, next.
- No clock. Close it mid-question and nothing is lost.
- Counter is scoped to the filter — 12 / 240 , not 12 / 18,651 .

TABLES USED

questions · papers · directions · attempt_answers

Build a set
How many [20 ▼]
Section [Reasoning ▼]
Source (•) Mixed
() My weak areas
[Start]

Q 7 of 20

⌚ 13:42

- A set that **ends** — that's what builds the daily habit.
- Timed, scored, with a per-section breakdown.
- “My weak areas” is the AI mode. Needs `topic`, which we don't have.

TABLES USED

everything the Bank uses + attempts + user_topic_stats

Past Papers		SBI ▼	P0 ▼
Mains			
2022	190 qs	3 hrs	►
2019	190 qs	3 hrs	✓ 142
Prelims			
2022	100 qs	1 hr	►

- Browsed by exam → stage → year, the way aspirants already think.
- The real thing: sectional timing, negative marking, score at the end.
- One paper per real exam — `is_canonical` hides the duplicates.

TABLES USED

```
everything above + papers.duration_min · section_timing · total_marks
```

Order to build in

#	BUILD	WAITING ON
1	The six tables	Nothing
2	Import script, dry-run first	Nothing
3	Question Bank	Nothing — answers make it useful, not possible
4	Answer keys into the data	Parser work
5	PYQ Mix	Step 4 + marks
6	Past Papers	Step 4 + durations
7	Section & topic labels	Unlocks weak areas and mastery

Steps 1–3 can start today. Everything from 5 onward waits on answer keys.

Counts from every file under `India/banking/papers-deduped/` at commit `0ff28ea` — 305 papers, 18,651 questions.